

Diramu Kana Godana

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EDUCATION

Harvard University

Cambridge, MA

A.B. Concentration in Computer Science and Economics

Aug 2023 – May 2027

- **Coursework:** Discrete Mathematics, Intro to Algorithms, Linear Algebra, Data Structures & Algorithms, Computation for Contemporary Sciences, Visualization, Introduction to Probability, GenAI
- **Clubs:** Harvard WECODE, Harvard Women in CS, Harvard Undergraduate Machine Intelligence Community (HUMIC), Harvard Society of Black Scientists and Engineers, Harvard African Students Association

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C/C++, SQL, HTML/CSS, JSON

Frameworks & Libraries: FastAPI, React.js, Next.js, LangChain, Tailwind CSS, Node.js, Bootstrap, Frappe, Flask, TensorFlow, PyTorch

Databases & Tools: MySQL, PostgreSQL, Pinecone (vector database), Git, Docker, REST APIs

RELEVANT WORK EXPERIENCE

Harvard College Research Program

Cambridge, MA

AI Research and Software Engineering Intern

June 2025 – Aug 2025

- Built a bilingual Retrieval-Augmented Generation (RAG) tutor for high schoolers, enabling Swahili-speaking students to learn curriculum-aligned material from English textbooks.
- Engineered document pipelines (PDF parsing, chunking, embeddings, metadata filters) with LangChain, OpenAI API, and Pinecone; processed 500+ textbook chunks with 95% retrieval accuracy.
- Developed three interactive tutor modes (summarization, revision Q&A, free-form questioning) with a Next.js frontend and FastAPI backend, achieving sub-2s average response times across 100+ queries.
- Innovated dual-language output (English + Swahili) to increase accessibility, guaranteeing curriculum-compliant and exam-oriented answers aligned with Kenya's 8-4-4 system; tested iteratively with 56 pilot students

Kerugoya County Referral Hospital

Kerugoya, Kenya

Software Development Engineering Intern / Priscilla Chan Summer Service Fellow

June 2024 – Aug 2024

- Designed and implemented a digital patient management system, automating records for over 1,000 patients.
- Developed ERP and EMR systems with Java, Python, and Frappe; used MySQL, REST APIs, and Docker.
- Participated in Agile cycles, conducting code reviews and testing for system reliability.

TECHNICAL PROJECTS (VIEW FULL PROJECT GALLERY HERE)

Adaptive Swahili-English AI Curriculum Tutor

June 2025 – Aug 2025

- Developed a bilingual Retrieval-Augmented Generation (RAG) tutor for Form 1 Biology/Geography, supporting summarization, revision Q&A, and free-form questioning.
- Engineered document pipelines (chunking, embeddings, metadata filtering) with LangChain, Pinecone, and GPT-4 Turbo to deliver curriculum-aligned answers.
- Built a responsive frontend in Next.js/React with Tailwind CSS and integrated a FastAPI backend for scalable deployment.

Multilingual Flashcard Generator

Mar 2025 – May 2025

- Designed and deployed a full-stack web app that generates interactive multilingual flashcards from uploaded .pdf, .docx, and .txt documents using GPT-4 for summarization and translation.
- Implemented a Flask backend to handle document parsing, API calls, and bilingual flashcard data flow; integrated OpenAI TTS for natural audio playback in both languages.
- Built a dynamic React frontend with Tailwind CSS featuring animated themes, flashcard flipping interactions, and session-based learning history.
- Enhanced user experience with document export, customizable themes, and multiple study modes.

Integrated Patient Management System

June 2024 – Aug 2024

- Led design of scheduling, billing, and analytics modules for a comprehensive patient management system.
- Employed Java for core functionality; integrated Python and Frappe for backend logic and UI workflows.

LEADERSHIP & ACTIVITIES

Harvard WECODE

Cambridge, MA

Associate Director of Engagement

Sept 2024 – Present

- Facilitated the largest student-run women's tech conference, held on Harvard's campus.
- Oversaw outreach and logistics, streamlining communication between speakers, sponsors, and attendees.

Harvard Undergraduate Machine Intelligence Community (HUMIC)

Cambridge, MA

Future of Intelligence Fellow (FIF)

Sept 2024 – Dec 2024

- Explored ML/AI and ethical AI frameworks through a selective semester- long fellowship.
- Trained CNNs for image and video interpretation tasks, refined prompting to enhance LLM performance.